

PHP & MySQL

What is MySQL?

MySQL is an open-source relational database management system (RDBMS). It is the most popular database system used with PHP. MySQL is developed, distributed, and supported by Oracle Corporation.

- The data in a MySQL database are stored in tables which consists of columns and rows.
- MySQL is a database system that runs on a server.
- MySQL is ideal for both small and large applications.
- MySQL is very fast, reliable, and easy to use database system. It uses standard SQL
- MySQL compiles on a number of platforms.

How to connect PHP with MySQL Database?

PHP 5 and later can work with a MySQL database using:

1. MySQLi extension.
2. PDO (PHP Data Objects).

PHP with MySQL Functions:-

Fncion	Description	PHP
<u>mysql_close()</u>	Closes a non-persistent MySQL connection	3
<u>mysql_affected_rows()</u>	Returns the number of affected rows in the previous MySQL operation	3
<u>mysql_connect()</u>	Opens a non-persistent MySQL connection	3
<u>mysql_fetch_array()</u>	Returns a row from a recordset as an associative array and/or a numeric array	3
<u>mysql_fetch_assoc()</u>	Returns a row from a recordset as an associative array	4
<u>mysql_fetch_row()</u>	Returns a row from a recordset as a numeric array	3
<u>mysql_num_rows()</u>	Returns the number of rows in a recordset	3
<u>mysql_query()</u>	Executes a query on a MySQL database	3
<u>mysql_select_db()</u>	Sets the active MySQL database	3

In Mysql following functions are available:-

- **mysqli_connect()**
- **mysqli_select_db ()**
- **mysqli_query ()**
- **mysqli_num_rows ()**
- **mysqli_fetch_array ()**
- **mysqli_affected_rows ()**
- **mysqli_close()**

1. Mysql_connect() or mysqli_connect()

- The PHP mysql connect function is used to connect to a MySQL database server.
- Syntax:
- **Mysql_connect:**
`$cn=mysql_connect(servername,username,password);`
- **Mysqli_connect:**
`$cn=mysqli_connect(servername,username,password);`
- **Example:**
`$cn=mysql_connect("localhost","root","");`

2. mysql_select_db () or mysqli_select_db ()

- The mysql_select_db function is used to select a database.
- It has the following syntax.
 - **mysql_select_db function :-**
`mysql_select_db($database_name,$ cn);`

- **mysqli_select_db function :-**

```
mysqli_select_db($cn,$database_name);
```

- **Example:**

```
$cn=mysqli_connect("localhost","root","");
```

```
$db=mysqli_select_db("student",$cn);
```

3. mysql_query() or mysqli_query ()

- The mysql_query function is used to execute [SQL](#) queries.
- The function can be used to execute the following query types;
 - Insert
 - Select
 - Update
 - delete

It has the following syntax.

- **mysql_query:-**

```
<?php
mysql_query($query,$cn) ;
?>
```

- **mysqli_query:-**

```
<?php
mysqli_query($cn,$query) ;
?>
```

- **Example:**

```
$cn=mysqli_connect("localhost","root","");
```

```
$db=mysqli_select_db("student",$cn);
```

```
$q=mysqli_query("select * from result",$cn);
```

4. **mysql_num_rows ()** or **mysqli_num_rows()**

- The **mysqli_num_rows** function is used to get the number of rows returned from a select query.
- It has the following syntax.
- **mysql_num_rows:-**

```
<?php
$str=mysql_num_rows($result);
?>
```

- **mysqli_num_rows:-**

```
<?php
$str= mysqli_num_rows($result);
?>
```

- here \$str is total no. of rows.

- **Example**

```
$cn=mysql_connect("localhost","root","");
$db=mysql_select_db("student",$cn);
$q=mysql_query("select * from result",$cn);

$str=mysql_num_rows($result);
```

5. **mysql_fetch_array()** or **mysqli_fetch_array ()**

- The **mysql_fetch_array()** function returns a row from a recordset as an associative array and/or a numeric array.
- It has the following syntax.

```
$row= mysql_fetch_array(data,array_type)
```

- Parameters:-

Parameter	Description
data	Required. Specifies which data pointer to use. The data pointer is the result from the mysql_query() function
array_type	Optional. Specifies what kind of array to return. Possible values: - MYSQL_ASSOC - Associative array - MYSQL_NUM - Numeric array - MYSQL_BOTH - Default. Both associative and numeric array

- Mysql_fetch_array()**

`mysql_fetch_array($result_of_Query)`

- Mysqli_fetch_array()**

`$row = mysqli_fetch_array($result_of_Query)`

- eg.**

```
sql = "SELECT * from Person WHERE Lastname='Sharma'";
$result = mysql_query($sql,$con);
print_r(mysql_fetch_array($result));
```

6. mysql_affected_rows() or mysqli_affected_rows ()

- The mysql_affected_rows() function returns the number of affected rows in the previous MySQL operation.
- This function returns the number of affected rows on success, or -1 if the last operation failed.

- **Mysql_affected_rows()**

```
$n=mysql_affected_rows()
```

- **Mysqli_ affected_rows()**

```
$n=mysqli_affected_rows($con);
```

- **Example-**

```
<?php

$con = mysql_connect("localhost","root","");

if (!$con)

{

    die("Could not connect: " . mysql_error());

}

mysql_select_db("mydb",$con);

mysql_query("DELETE FROM mytable WHERE id < 5");

$rc = mysql_affected_rows();

echo "Records deleted: " . $rc;

mysql_close($con);

?>
```

7. `mysql_close()` or `mysqli_close()`

- `mysql_close()` Function is used to close an existing opened database connection between PHP scripting and MySQL Database Server.
- **`Mysql_close()`**

`mysql_close("Connection Object");`

eg. `mysql_close($con);`

- **`Mysqli_close()`**

`mysqli_close("Connection Object");`

eg. `mysqli_close($con);`